Project Design Phase-II Technology Stack (Architecture & Stack)			
Date	31 August 2026	**Team ID / Scope**	x_1959081_emp_issue
Project Name	Employee Raise Issue Integration	**Maximum Marks**	4 Marks

TECHNICAL ARCHITECTURE

The system architecture illustrates the end-to-end multi-tier flow from end-user interactions on the Service Portal down to the ServiceNow application logic engine, custom relational data store, and platform API integrations.

END-TO-END SYSTEM ARCHITECTURE (3-TIER CLOUD DEPLOYMENT)

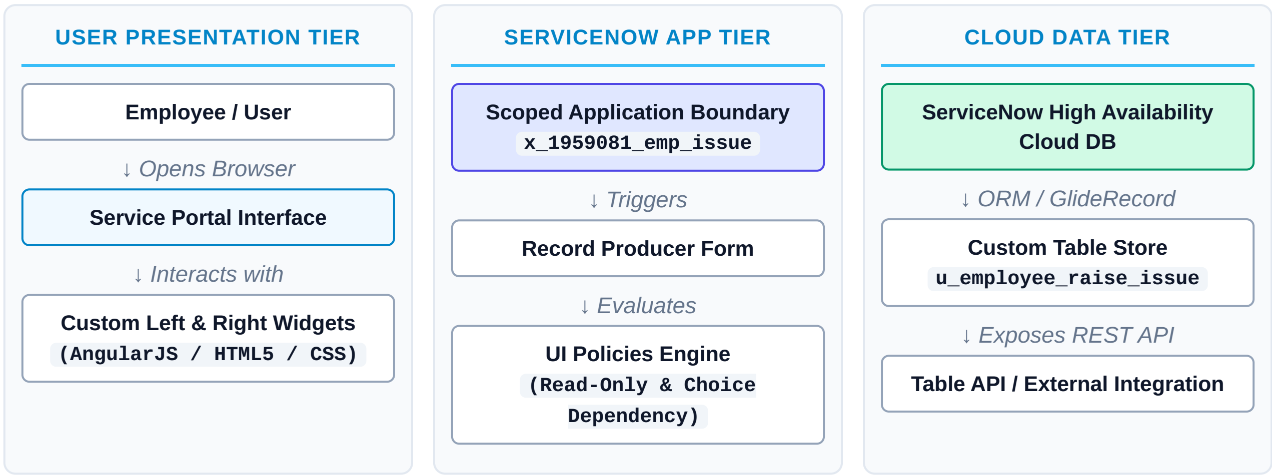

TABLE-1 : COMPONENTS & TECHNOLOGIES

S.No	Component	Description	Technology
1	**User Interface**	How user interacts with application e.g. Service Portal, Custom Left & Right Widgets.	HTML5, CSS3, Bootstrap 3, AngularJS, Service Portal Framework
2	**Application Logic-1**	Logic for dynamic form field control, read-only protection, and parent-child choice dependencies.	JavaScript (ES6), Client Scripts, ServiceNow UI Policies Engine
3	**Application Logic-2**	Logic for front-end form data transformation and automated table record generation.	ServiceNow Record Producer Scripting, JavaScript API (GlideRecord / GlideSystem)
4	**Application Logic-3**	Logic for application scope isolation, component encapsulation, and namespace resolution.	ServiceNow Scoped Application Framework (`x_1959081_emp_issue`)
5	**Database**	Custom relational database table engineered to capture issue metadata, categories, sub-categories, and user references.	ServiceNow MariaDB / MySQL Relational Engine (Custom Table)
6	**Cloud Database**	Database Service on Cloud backing the ServiceNow multi-tenant infrastructure.	ServiceNow High Availability Cloud DB (Multi-Node Replicated)
7	**File Storage**	File storage requirements for ticket attachments, portal widget assets, and static media.	ServiceNow `sys_attachment` Table & Content Storage Engine
8	**External API-1**	Purpose of External API used in the application for ticket status querying and external integrations.	ServiceNow Table API / REST API Framework (JSON over HTTPS)
9	**Infrastructure (Server / Cloud)**	Application Deployment on Cloud Platform Instance. Cloud Server Configuration: San Diego / Utah / Washington Instance	ServiceNow Multi-Tenant Cloud Infrastructure

TABLE-2: APPLICATION CHARACTERISTICS

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	Client-side rendering frameworks powering custom Service Portal widgets and responsive layout grids.	AngularJS v1.x, Bootstrap 3 Grid Framework, FontAwesome Icons
2	**Security Implementations**	Role-Based Access Control (RBAC), scoped application isolation, and read-only ticket protection policies.	ServiceNow ACLs, Application Scope Rules, HTTPS / TLS 1.3
3	**Scalable Architecture**	3-tier architecture separating Presentation Portal, Application Engine, and Cloud Relational DB.	ServiceNow 3-Tier Multi-Instance Cloud Architecture
4	**Availability**	High availability deployment with multi-node redundant servers and load balancers.	ServiceNow Advanced High Availability (AHA) Multi-Node Architecture
5	**Performance**	Design considerations for performance (client-side script optimization, lazy loading, and database indexing).	Client-Side Caching, Query Indexing, Asynchronous GlideAjax Engine

REFERENCES

- https://c4model.com/
- https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
- https://www.ibm.com/cloud/architecture
- https://aws.amazon.com/architecture
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